HW5

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```
# set seed
set.seed(123)
```

Question 1

Jeffrey's (1961) suggested a default rule for generating a prior distribution of a parameter θ in a sampling model $p(y \mid \theta)$. Jeffreys' prior is given by

$$p_J(\theta) \propto \sqrt{I(\theta)},$$

where

$$I(\theta) = -E\left[\frac{\partial^2 \log p(Y \mid \theta)}{\partial \theta^2} \mid \theta\right]$$

is the Fisher information.

Part A

Let $Y \sim$ Binomial(n, θ). *Show that* $p_J(\theta) \propto Beta(1/2, 1/2)$.

We have that

$$p_J(\theta) \propto \sqrt{I(\theta)},$$

where

$$I(\theta) = -E\left[\frac{\partial^2 \log p(Y \mid \theta)}{\partial \theta^2} \mid \theta\right]$$

is the Fisher information. We want to show that $p_J(\theta) \propto Beta(1/2, 1/2)$. Let $Y \sim$ Binomial(n, θ), so we have $p(Y|\theta) = \binom{n}{y}\theta^y * (1-\theta)^{n-y}$ First, we will simplify $\frac{\partial^2}{\partial \theta^2} \log(p(Y|\theta))$:

$$\begin{aligned}
\frac{\partial^2}{\partial\theta^2}\log(p(Y|\theta)) &= \frac{\partial^2}{\partial\theta^2}\log[\binom{n}{y}\theta^y * (1-\theta)^{n-y}] \\
&= \frac{\partial^2}{\partial\theta^2}\log[\binom{n}{y}] + y * \log[\theta] + (n-y)\log[(1-\theta)] \\
&= \frac{\partial}{\partial\theta}\left(\frac{\partial}{\partial\theta}\left(\log[\binom{n}{y}] + y * \log[\theta] + (n-y)\log[(1-\theta)]\right)\right) \\
&= \frac{\partial}{\partial\theta}\left(y * \frac{1}{\theta} + (n-y)\frac{1}{(1-\theta)}\right) \\
&= -y * \frac{1}{\theta^2} - (n-y)\frac{1}{(1-\theta)^2}.
\end{aligned}$$

Then, taking the expectation of the simplified expression yields

$$\begin{aligned}
\mathbf{E}[-y * \frac{1}{\theta^2} - (n-y)\frac{1}{(1-\theta)^2}] &= -\frac{1}{\theta^2}\mathbf{E}[y] - \frac{1}{(1-\theta)^2}\mathbf{E}[(n-y)]] \\
&= -\frac{1}{\theta^2}\mathbf{E}[y] - \frac{1}{(1-\theta)^2}(n - \mathbf{E}[y])
\end{aligned}$$

Since Y is sampled from a binomial distribution, we know $\mathbf{E} = n * \theta$. Observe

$$\begin{aligned}
-\frac{1}{\theta^2}\mathbf{E}[y] - \frac{1}{(1-\theta)^2}(n - \mathbf{E}[y]) &= -\frac{1}{\theta^2} * (n\theta) - \frac{1}{(1-\theta)^2}(n - (n\theta)) \\
&= -\frac{n}{\theta} - \frac{n(1-\theta)}{(1-\theta)^2} \\
&= -\frac{n}{\theta} - \frac{n(1-\theta)}{(1-\theta)^2} \\
&= -\frac{n}{\theta} - \frac{n}{(1-\theta)} \\
&= -\frac{n(1-\theta)}{\theta * (1-\theta)} - \frac{n * \theta}{\theta * (1-\theta)} \\
&= -\frac{-n + n * \theta - n * \theta}{\theta * (1-\theta)} \\
&= -\frac{-n}{\theta * (1-\theta)} \\
&\propto \theta^{-1/2} * (1-\theta)^{-1/2} \\
&= Beta(1/2, 1/2).
\end{aligned}$$

Therefore, $\boxed{p_J(\theta) \propto Beta(1/2, 1/2)}$.

Part B

Re-parameterize the model in part a such that $\psi = \log[\theta/(1-\theta)]$. *This implies that* $p(y \mid \psi) = \binom{n}{y} e^{\psi y}(1+e^{\psi})^{-n}$. *Show that*

$$p_J(\psi) \propto \frac{n^{1/2} e^{\psi/2}}{1+e^{\psi}} \propto \frac{e^{\psi/2}}{1+e^{\psi}}.$$

Re-parameterizing the model in part a such that $\psi = \log[\theta/(1-\theta)]$ now gives $p(y|\psi) = \binom{n}{y} e^{\psi y} * (1+e^{\psi})^{-n}$. Observe

$$p_J(\psi) \propto \sqrt{I(\psi)} = \sqrt{-\mathbf{E}\left[\frac{\partial^2 \log p(Y \mid \psi)}{\partial \psi^2} \mid \psi\right]}.$$

First, consider the following:

$$\begin{aligned}
\frac{\partial^2 \log p(y \mid \psi)}{\partial \psi^2} &= \frac{\partial^2}{\partial \psi^2} \log\left(\binom{n}{y} e^{\psi y}(1+e^{\psi})^{-n}\right) \\
&= \frac{\partial^2}{\partial \psi^2}\left[\log\left(\binom{n}{y}\right) + \psi y - n\log(1+e^{\psi})\right] \\
&= \frac{\partial}{\partial \psi}\left[\frac{\partial}{\partial \psi}\log\left(\binom{n}{y}\right) + y - n\frac{e^{\psi}}{1+e^{\psi}}\right] \\
&= \frac{\partial}{\partial \psi}\left[y - ne^{\psi}(1+e^{\psi})^{-1}\right] \\
&= -ne^{\psi}(1+e^{\psi})^{-1} + ne^{2\psi}(1+e^{\psi})^{-2} \\
&= -ne^{\psi}(1+e^{\psi})^{-1}[1 - e^{\psi}(1+e^{\psi})^{-1}] \\
&= -ne^{\psi}(1+e^{\psi})^{-1}[\frac{(1+e^{\psi}) - e^{\psi}}{(1+e^{\psi})}] \\
&= -ne^{\psi}(1+e^{\psi})^{-1}[\frac{1}{(1+e^{\psi})}] \\
&= -\frac{ne^{\psi}}{(1+e^{\psi})^2}
\end{aligned}$$

Now, substituting the above expression into

$$p_J(\psi) \propto \sqrt{I(\psi)} = \sqrt{-\mathbf{E}\left[\frac{\partial^2 \log p(Y \mid \psi)}{\partial \psi^2} \mid \psi\right]}.$$

yields the following:

$$\sqrt{-\mathbf{E}\left[\frac{\partial^2 \log p(Y \mid \psi)}{\partial \psi^2} \mid \psi\right]} = \sqrt{-\mathbf{E}\left[-\frac{ne^{\psi}}{(1+e^{\psi})^2}\right]}$$
$$= \frac{n^{1/2} e^{\psi/2}}{1+e^{\psi}}$$

Therefore, we have

$$p_J(\psi) \propto \frac{n^{1/2} e^{\psi/2}}{1+e^{\psi}} \propto \frac{e^{\psi/2}}{1+e^{\psi}}.$$

Part C

Take the prior distribution from (a) and apply the change of variables formula from exercise 3.10 (see Homework 4) to obtain the induced prior density on ψ. This density should be the same as the one derived in part b) of this exercise. This consistency under re-parametrization is the defining characteristic of Jeffereys' prior.

The prior distribution from a) is $p_J(\theta) \propto Beta(1/2, 1/2) \propto \theta^{-1/2}(1-\theta)^{-1/2}$. Using the change of variables formula from exercise 3.10, we have $h(\psi) = \theta = \frac{e^{\psi}}{1+e\psi}$ and $p_J(\psi) = p_J(h(\psi)) \times \left|\frac{dh}{d\psi}\right|$. We know $p_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2}$. Observe

$$p_J(h(\psi)) = p_J(\frac{e^{\psi}}{1+e\psi}) \propto \left(\frac{e^{\psi}}{1+e\psi}\right)^{-1/2} * \left(1 - \frac{e^{\psi}}{1+e\psi}\right)^{-1/2}$$
$$= \frac{1+e^{\psi}}{e^{\psi/2}}.$$

Now, observe:

$$\left|\frac{dh}{d\psi}\right| = \left|\frac{d}{d\psi}\frac{e^{\psi}}{1+e^{\psi}}\right|$$
$$= \frac{(1+e^{\psi})e^{\psi} - e^{\psi}e^{\psi}}{(1+e^{\psi})^2}$$
$$= \frac{e^{\psi}(1+e^{\psi}-e^{\psi})}{(1+e^{\psi})^2}$$
$$= \frac{e^{\psi}}{(1+e^{\psi})^2}.$$

Thus, we obtained

$$\begin{aligned} p_J(\psi) &= p_J(h(\psi)) \times \left|\frac{dh}{d\psi}\right| \\ &= \left(\frac{1+e^{\psi}}{e^{\psi/2}}\right)\left(\frac{e^{\psi}}{(1+e^{\psi})^2}\right) \\ &= \frac{e^{\psi/2}}{1+e^{\psi}}. \end{aligned}$$

This is the same density we derived in part b) of this exercise, as desired.

I used ChatGPT to re-format my long equations. The steps are my own.

Question 2

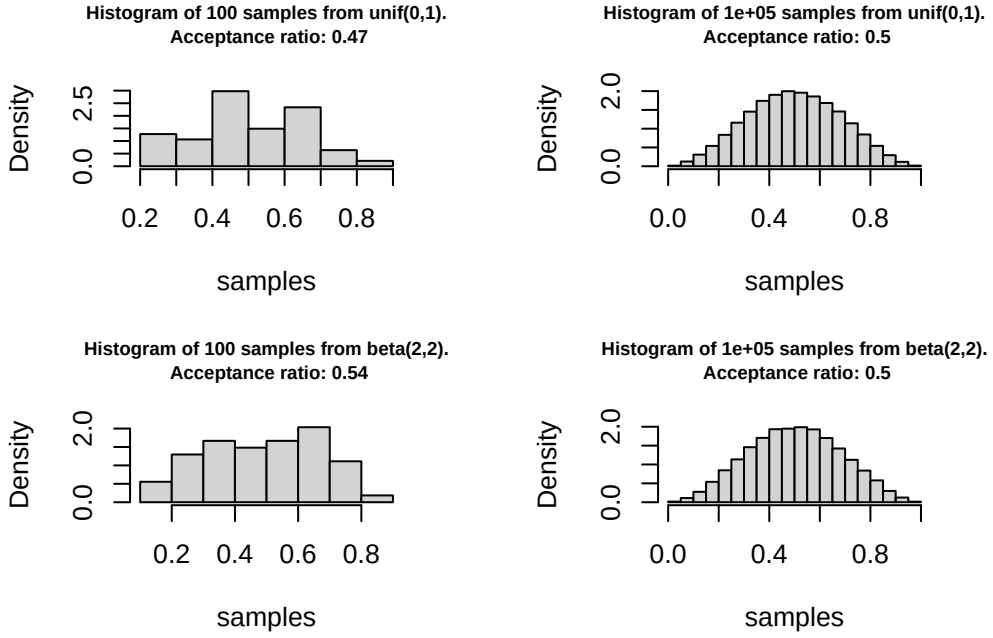
Part A

Provide the four histograms from Tasks 2 and 3 using the Uniform(0,1) and the Beta(2,2) enveloping proposals. Provide the acceptance ratios. Provide commentary.

```
# grid of points
x <- seq(0, 1, 10^-2)
fx <- function(x) sin(pi * x)^2
sim_fun <- function(f, envelope = "unif", par1 = 0, par2 = 1, n = 10^2, plot = TRUE){

  r_envelope <- match.fun(paste0("r", envelope))
  d_envelope <- match.fun(paste0("d", envelope))
  proposal <- r_envelope(n, par1, par2)
  density_ratio <- f(proposal) / d_envelope(proposal, par1, par2)
  samples <- proposal[runif(n) < density_ratio]
  acceptance_ratio <- length(samples) / n
  if (plot) {
    hist(samples, probability = TRUE,
         main = paste0("Histogram of ",
                       n, " samples from ",
                       envelope, "(", par1, ",", par2,
                       ").\n Acceptance ratio: ",
                       round(acceptance_ratio,2)),
         cex.main = 0.75)
  }
  list(x = samples, acceptance_ratio = acceptance_ratio)
}
par(mfrow = c(2,2), mar = rep(4, 4))
```

```r
unif_1 <- sim_fun(fx, envelope = "unif", par1 = 0, par2 = 1, n = 10^2)
unif_2 <- sim_fun(fx, envelope = "unif", par1 = 0, par2 = 1, n = 10^5)
beta_1 <- sim_fun(fx, envelope = "beta", par1 = 2, par2 = 2, n = 10^2)
beta_2 <- sim_fun(fx, envelope = "beta", par1 = 2, par2 = 2, n = 10^5)
```




```r
par(mfrow = c(1,1))
```

Here, we provided the four histograms from Tasks 2 and 3 using the Uniform(0,1) and the Beta(2,2) enveloping proposals, along with acceptance ratios. We obtained similar acceptance ratios between 0.47 and 0.54 across all 4 conditions. The simulations of size 10^5 show smoother histograms. Thus, it is not always the case that a larger simulation size (or an enveloping function that appears most simialr to the target function at first glance) will produce a higher (better) acceptance ratio.

Part B

(i) Do you recommend the Uniform(0,1) or the Beta(2,2) as a better enveloping function (or are they about the same)?

Since the acceptance ratios range between 0.47 - 0.54, both the Uniform(0,1) and Beta (2,2) perform similarly for this target function, so I recommend both based on the observed acceptance ratios.

(ii) If you were to try and find an enveloping function that had a high acceptance ratio, which one would you try and why? For part (ii), either back up your choice using a small paragraph or (ii) empirically illustrate that it's better. (If you wanted to go above and beyond, you could do both, which is highly encouraged and would really prepare you for the exam!)

To try and find an enveloping function with a high acceptance ratio, I would try some additional Beta(a,b) distributions that more closely resemble the target function, as discussed in Lab 05. I would then draw samples from that Beta distribution until I obtained a higher acceptance ratio. Although the Beta and Uniform enveloping functions have similar acceptance ratios in this case, we may select a Beta distribution with lower variance to more closely resemble the target function, leaving less area for "rejected" samples while possibly resulting in more rejections near the "tails" of the target distribution.

Alternatively, to try and find an enveloping function with a high acceptance ratio, I would choose a function that closely resembles the target function. Before using this function, we should make sure is relatively easy to sample from. Since $\text{eff}(q(x)) = \frac{1}{c}\int_x \tilde{f}(x)dx$ and here we have $\int_0^1 \sin^2(\pi x)dx = 0.5$. Here, the ideal enveloping function is $q(x) = 2 * \sin^2(\pi x)$ since this function would give $\tilde{f}(x) = \frac{1}{2}q(x)$, resulting in $\hat{c} = 1/2$ and a resulting acceptance ratio of 1:

$$\frac{1}{(1/2)}\int_0^1 \sin^2(\pi x)dx = 2 * 1/2 = 1.$$